Author: Anne Sabourin Description: DAMEX tutorial
(`unsupervised.feature_clustering' subpackage, damex` submodule)

Damex tutorial

This notebook implements the DAMEX algorithm as described in [1]. The unsupervised task considered is the discovery of groups of components of a random vector that are comparatively likely to be simultaneously large.

The tutorial introduces new methods for selecting the tuning parameter ϵ in DAMEX and κ in CLEF, primarily based on AIC and cross-validation. These methods involve a specific pseudo-likelihood for random subsets of features among $\{1, \ldots, d\}$. Although these methods lack theoretical investigation, this notebook provides empirical evidence supporting the relevance of the proposed selection criteria.

References: [1] Goix, N., Sabourin, A., & Clémençon, S. (2017). Sparse representation of multivariate extremes with applications to anomaly detection. Journal of Multivariate Analysis, 161, 12-31.

In [ ]:
```
import numpy as np
import matplotlib.pyplot as plt
import pprint as pp
import MLExtreme as mlx


# Define the norm function as the infinite norm (other norms are not
# implemented in DAMEX / CLEF). This definition is not required for
# running CLEF/DAMEX and is used here solely for visualization and
# analysis of the output.

def norm_func(x):
    return np.max(x, axis=1)
```

Ground truth definition:

Define or generate a list of subfaces of the unit sphere that represent the support of the limit measure.

In [ ]:
```
Plot = False
seed = 42
dim = 20  # try 5, 20, 50, 100
num_subfaces = 10  # try 5, 20, 50
subfaces_list = mlx.gen_subfaces(dimension=dim,
                                 num_subfaces=num_subfaces,
                                 max_size=10,  # try 4, 10, 20
                                 prevent_inclusions=True,
```

```
                                    seed=seed)

# # Uncomment for a simpler example:
# subfaces_list = [[0, 1], [1, 2], [2, 3, 4]]

if False:   # change to True to print the list of subfaces
    print(subfaces_list)
    pp.pprint(mlx.list_to_dict_size(subfaces_list))

subfaces_matrix = mlx.subfaces_list_to_matrix(subfaces_list, dim)
# print(subfaces_matrix) Dimension, number of mixture components,
# weights and Dirichlet center locations
n = int(np.sqrt(dim) * 10**3)
k = np.shape(subfaces_matrix)[0]
dim = np.shape(subfaces_matrix)[1]
# Define admissible Dirichlet mixture parameters for the limit angular measure
# of a marginally Pareto-standardized heavy-tailed vector,
# based on the matrix of subfaces.
# This avoids potentially imbalanced settings across features.
wei = np.ones(k)/k
Mu, wei = mlx.normalize_param_dirimix(subfaces_matrix, wei)
# Print the subfaces and their weights, and record them as a list for
# further use.
faces_true = subfaces_list
wei_true = wei
print(f'Mu matrix: \n {np.round(Mu, 3)}')
print(f'Weights: {np.round(wei, 3)}')
```

```
Mu matrix:
 [[0.3   0.    0.    0.4   0.    0.    0.    0.    0.3   0.    0.    0.
  0.    0.    0.    0.    0.    0.    0.    0.   ]
 [0.077 0.103 0.103 0.103 0.103 0.    0.103 0.077 0.    0.103 0.    0.
  0.154 0.    0.    0.    0.    0.    0.    0.077]
 [0.12  0.    0.    0.    0.    0.    0.16  0.12  0.    0.    0.    0.24
  0.    0.    0.    0.    0.24  0.    0.    0.12 ]
 [0.    0.    0.    0.    0.    0.    0.    0.136 0.136 0.    0.    0.
  0.    0.182 0.273 0.    0.    0.273 0.    0.   ]
 [0.2   0.    0.    0.    0.    0.8   0.    0.    0.    0.    0.    0.
  0.    0.    0.    0.    0.    0.    0.    0.   ]
 [0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.75  0.
  0.    0.25  0.    0.    0.    0.    0.    0.   ]
 [0.    0.    0.    0.    0.571 0.    0.    0.    0.429 0.    0.    0.
  0.    0.    0.    0.    0.    0.    0.    0.   ]
 [0.    0.2   0.    0.2   0.    0.    0.    0.    0.15  0.    0.    0.
  0.    0.    0.3   0.    0.    0.    0.    0.15 ]
 [0.    0.    0.125 0.    0.    0.    0.    0.    0.    0.125 0.    0.188
  0.188 0.    0.    0.    0.188 0.188 0.    0.   ]
 [0.    0.074 0.074 0.    0.074 0.    0.074 0.056 0.    0.074 0.    0.
  0.    0.074 0.    0.222 0.    0.    0.222 0.056]]
Weights: [0.042 0.162 0.104 0.092 0.062 0.067 0.029 0.083 0.133 0.225]
```

Difficulty Settings for the Tail Problem

The `lnu` parameter below is the logarithm of the concentration parameter for the Dirichlet mixture. It is a crucial parameter that determines the difficulty of the clustering problem. If

any $\exp(\mathrm{lnu}[i]) \times \mathrm{Mu}[i,j] < 1$, the problem is pathologically hard, as the mass on any subface concentrates on the boundary of that face. Conversely, if all $\exp(\mathrm{lnu}[i]) \times \mathrm{Mu}[i,j] \gg 1$, the problem is very easy.

Experiment with the 'difficulty' parameter below, which should be strictly positive. Values greater than one correspond to very hard problems, while values close to zero generate very easy problems. Interesting results occur for difficulty ≈ 0.8.

In [ ]:
```
difficulty = 0.8

min_mus = np.zeros(k)
for j in range(k):
    min_mus[j] = np.min(Mu[j, Mu[j, :] > 0])
lnu = np.log(1/(difficulty**2) * 1/min_mus)
# check (change to True  to display)
if False:
    print("absolute dirichlet parameters: ")
    print(Mu * np.exp(lnu).reshape(-1, 1))
```

Other parameter settings that govern the speed of convergence of the conditional distribution above a radial threshold towards the limit measure.

In [ ]:
```
# Regular variation index of the data
alpha = 2

# Centers of mass of the Dirichlet mixture in the bulk (vanishing
# impact above large radial thresholds):
Mu_bulk = np.ones((k, dim))/dim
```

`index_weight_noise` below indicates how fast the impact of noise decreases with large radial thresholds. Specifically, the noise's weight decreases as: $(C/\mathrm{radius})^{\mathrm{index_weight_noise}}$

In [ ]:
```
index_weight_noise = 4
```

Dataset Generation and Visualisation

In [ ]:
```
# Generate data
np.random.seed(42)
X = mlx.gen_rv_dirimix(alpha, np.round(Mu, 3),  wei, lnu,
                       scale_weight_noise=1, Mu_bulk=Mu_bulk,
                       index_weight_noise=index_weight_noise, size=n)


# Define rank-transformed data
Xt = mlx.rank_transform(X)

# Generate test data for unsupervised evaluation
np.random.seed(12345)
Xtest = mlx.gen_rv_dirimix(alpha, np.round(Mu, 3),  wei, lnu,
                           scale_weight_noise=1, Mu_bulk=Mu_bulk,
                           index_weight_noise=index_weight_noise, size=5*n)
```

```
# Define rank-transformed test data
std_Xtest = mlx.rank_transform(Xtest)
```

In [ ]:
```
# pairwise plot of generated data for two dependent components
len_faces = np.sum(subfaces_matrix, axis=1)
r_i = np.where(len_faces > 1)[0][0]
iplot = subfaces_list[r_i][0]
jplot = subfaces_list[r_i][1]

if Plot:
    plt.figure(figsize=(10, 10))
    X_disp = X**(alpha/4)  # for easier visualization only
    max_val = np.max(X_disp)
    scatter = plt.scatter(X_disp[:, iplot], X_disp[:, jplot], alpha=0.5)
    plt.xlim(0, max_val)
    plt.ylim(0, max_val)
    plt.xlabel('Feature 1')
    plt.ylabel('Feature d')
    plt.title('Scatter Plot of the raw dataset')
    plt.show()
```

In [ ]:
```
# Rank transformed  generated data with  unit pareto margins
# and visualization
if Plot:
    plt.figure(figsize=(10, 10))
    X_disp = Xt**(1/4)  # for easier viz only
    max_val = np.max(X_disp)
    scatter = plt.scatter(X_disp[:, iplot], X_disp[:, jplot], alpha=0.5)
    plt.xlim(0, max_val)
    plt.ylim(0, max_val)
    plt.xlabel('Feature 1')
    plt.ylabel('Feature d')
    plt.title('Scatter Plot of the rank transformed dataset')
    plt.show()
```

Data Analysis

Radial threshold selection before model fitting

In [ ]:
```
# Select the radial threshold via distance-covariance tests.
# See the tutorials on classification and regression for more details.
ntests_thresh = 10
ratio_ext = np.geomspace(0.05, 0.3, num=ntests_thresh)
pval, ratio_max = mlx.test_indep_radius_rest(Xt, y=None, ratio_ext=ratio_ext,
                                             norm_func=norm_func)

if Plot:
    mlx.plot_indep_radius_rest(pval, ratio_ext, ratio_max, n)

print(f'maximum ratio of extreme samples from rule-of-thumb \
distance covariance test: {mlx.round_signif(ratio_max, 2)}')
```

```
ratio_extremes = ratio_max * 4/5
norm_Xt = norm_func(Xt)
# Radial threshold selected with rule-of-thumb radius-versus-rest
# independence tests:
threshold = np.quantile(norm_Xt, 1 - ratio_extremes)
# Note: The following is not 'k' in the notations of Goix et al.'s
# paper.  Instead, it is between k and d * k. It is the number of data
# points above the radial threshold.
number_extremes = np.sum(norm_Xt >= threshold)

# %%%%
```

```
                                        damex_masses)

faces_dict_true, mass_dict_true = mlx.list_to_dict(faces_true, wei_true)

if True:
    print("List of subfaces found by DAMEX:")
    pp.pprint(faces_dict)
    print("(Damex) Associated limit mass:")
    print(mass_dict)
    print("True  list of subfaces: ")
    pp.pprint(faces_dict_true)
    print("(True) Associated limit mass:")
    print(mass_dict_true)
```

```
List of subfaces found by DAMEX:
{1: [],
 2: [[10, 13], [0, 5], [4, 8]],
 3: [[0, 3, 8]],
 4: [[7, 13, 14, 17]],
 5: [[7, 8, 13, 14, 17],
     [1, 3, 8, 14, 19],
     [9, 11, 12, 16, 17],
     [2, 11, 12, 16, 17]],
 6: [[2, 9, 11, 12, 16, 17], [0, 6, 7, 11, 16, 19]],
 7: [],
 8: [],
 9: [],
 10: [[0, 1, 2, 3, 4, 6, 7, 9, 12, 19]]}
(Damex) Associated limit mass:
{1: [], 2: [0.97, 0.95, 0.82], 3: [0.66], 4: [0.18], 5: [0.68, 0.58, 0.22, 0.2], 6:
[0.62, 0.31], 7: [], 8: [], 9: [], 10: [0.26]}
True  list of subfaces:
{1: [],
 2: [[0, 5], [10, 13], [4, 8]],
 3: [[0, 3, 8]],
 4: [],
 5: [[7, 8, 13, 14, 17], [1, 3, 8, 14, 19]],
 6: [[0, 6, 7, 11, 16, 19], [2, 9, 11, 12, 16, 17]],
 7: [],
 8: [],
 9: [],
 10: [[0, 1, 2, 3, 4, 6, 7, 9, 12, 19], [1, 2, 4, 6, 7, 9, 13, 15, 18, 19]]}
(True) Associated limit mass:
{1: [], 2: [0.062, 0.067, 0.029], 3: [0.042], 4: [], 5: [0.092, 0.083], 6: [0.1, 0.1
3], 7: [], 8: [], 9: [], 10: [0.16, 0.22]}
```

Unsupervised scoring

This material presents novel concepts, not yet supported by formal published theory, but intuitive and practical. It aids in evaluating goodness-of-fit, and most importantly, selecting the parameter `epsilon` (or `kappa_min` in CLEF). The ideas are related to Jorgensen's *Theory of Dispersion Models*.

We introduce an unsupervised performance metric termed *Total Deviance*, a denomination

inspired Jorgensen's work (see This metric relies on a unit deviance function $d(\text{subface}_1, \text{subface}_2)$, defined as in Chiapino et al. as the ratio of the symmetric difference between the index sets of two subfaces to the cardinality of the union of these index sets.

Given a unit deviance function d, the total deviance of an estimated pair (list of subfaces, list of masses) from Damex or CLEF, with respect to an extreme dataset, is computed as follows:

1. Each extreme point X_i is transformed into a binary vector using the same "mask" as in Damex/CLEF. For Damex, $X_{i,j} = 1$ if $||X_i|| > \text{threshold}$ and $X_{i,j} > \epsilon \times \text{threshold}$; otherwise, $X_{i,j} = 0$. In CLEF, this condition applies with $\epsilon = 1$.

2. Each binary-transformed extreme point is identified with a subface by considering its non-zero entries as the indices defining the subface. This allows the definition of $d(\text{subface}, X_i)$.

3. If $(m_j, j \leq J)$ are the estimated subface masses from Damex, their weights are defined as $p_j = m_j / \text{total_mass}$, where `total_mass` is the total limit measure mass on the complementary set of the unit ball.

The total (log-)deviance is then defined as:

$$\text{Deviance}(\text{subfaces}, \text{masses}, \text{data}) = \frac{-2}{n_{\text{extremes}}} \times \sum_{i \leq n_{\text{extremes}}} \log\left(\sum_{j \leq J} p_j \exp(-\text{rate} \times d(X_i, \text{su}\right.$$

Using this deviance, an AIC criterion is defined as:

$$\text{AIC}(\text{subfaces}, \text{masses}, \text{training_data}) = \text{Deviance}(\text{subfaces}, \text{masses}, \text{training_data}) +$$

This represents the AIC divided by the sample size.

Additionally, a cross-validation estimate of the Deviance (evaluated on a test set) is implemented. Both the AIC and cross-validation estimator aim to estimate the expected Deviance of the trained model on a test set.

Additionally, since we are working with simulated data, the implementations of Damex and CLEF include an option to compute the deviance of the estimated parameters (list of subfaces and list of masses) with respect to the true parameters (list of faces and list of weights), and vice versa.

References

[1] Cordeiro, G. M., Labouriau, R., & Botter, D. A. (2021).
An introduction to Bent Jørgensen's ideas. Brazilian Journal of Probability and Statistics, 35(1), 2-20.

[2] Jørgensen, B. (1987). Exponential dispersion models.
Journal of the Royal Statistical Society Series B: Statistical Methodology,
49(2), 127-145.

[3] Jorgensen, B. (1997). The theory of dispersion models. CRC Press.

In [ ]:
```
# Deviance from estimated to true parameters:
deviance_est_true, deviance_true_est = clustering.deviance_to_true(
    faces_true, wei_true)


print("AIC deviance estimate of expected deviance:")
print(clustering.get_AIC(Xt, standardize=False))

print("CV estimate of expected deviance")
print(np.mean(clustering.deviance_CV(Xt, standardize=False)))
print("test set  estimate of expected deviance")
print(clustering.deviance(std_Xtest, standardize=False))

print("Deviances between faces, masses: \
a) estimated to true b) true to estimated")
print(deviance_est_true, deviance_true_est)
```

```
AIC deviance estimate of expected deviance:
8.156902341065242
CV estimate of expected deviance
8.206601861485684
test set  estimate of expected deviance
8.202703587364631
Deviances between faces, masses: a) estimated to true b) true to estimated
5.6007403294885565 7.260997498963659
```

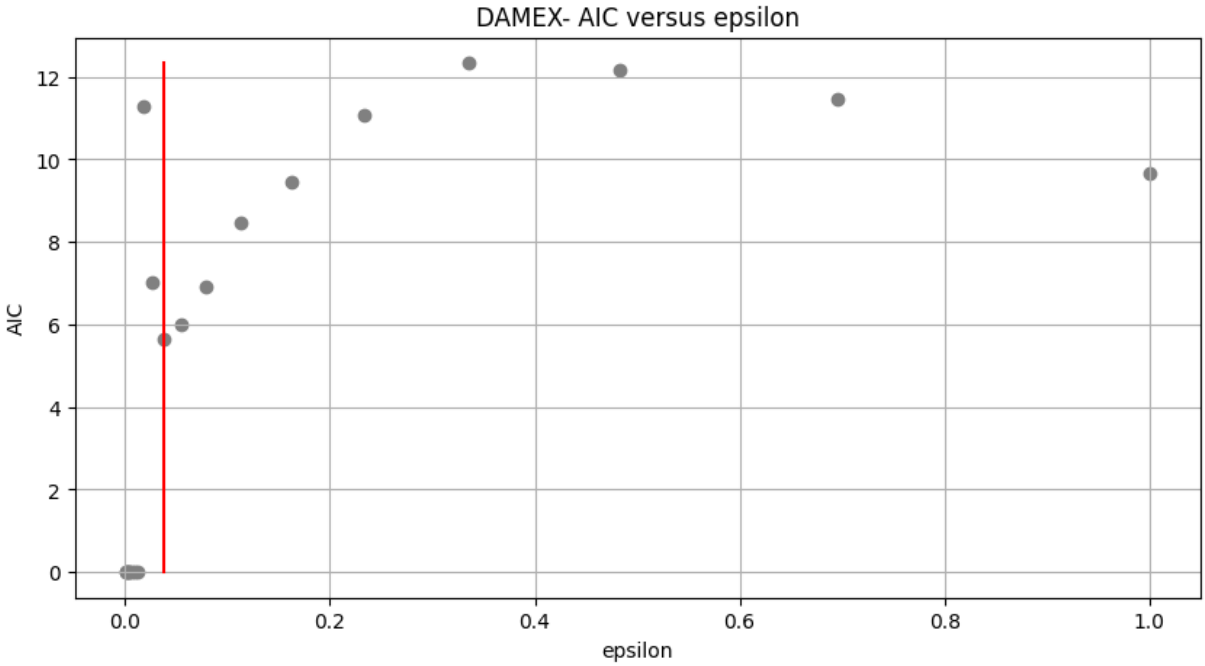
For this sample size, AIC and CV appear to be equally reliable for estimating the expected deviance on a test set (or the deviance on a sufficiently large test set as here). Additionally, the deviance between the true parameters (faces and masses) and the estimated ones is, in principle, similar to the test set estimate of the total deviance, which is also observed here.

Choosing Epsilon based on AIC / CV

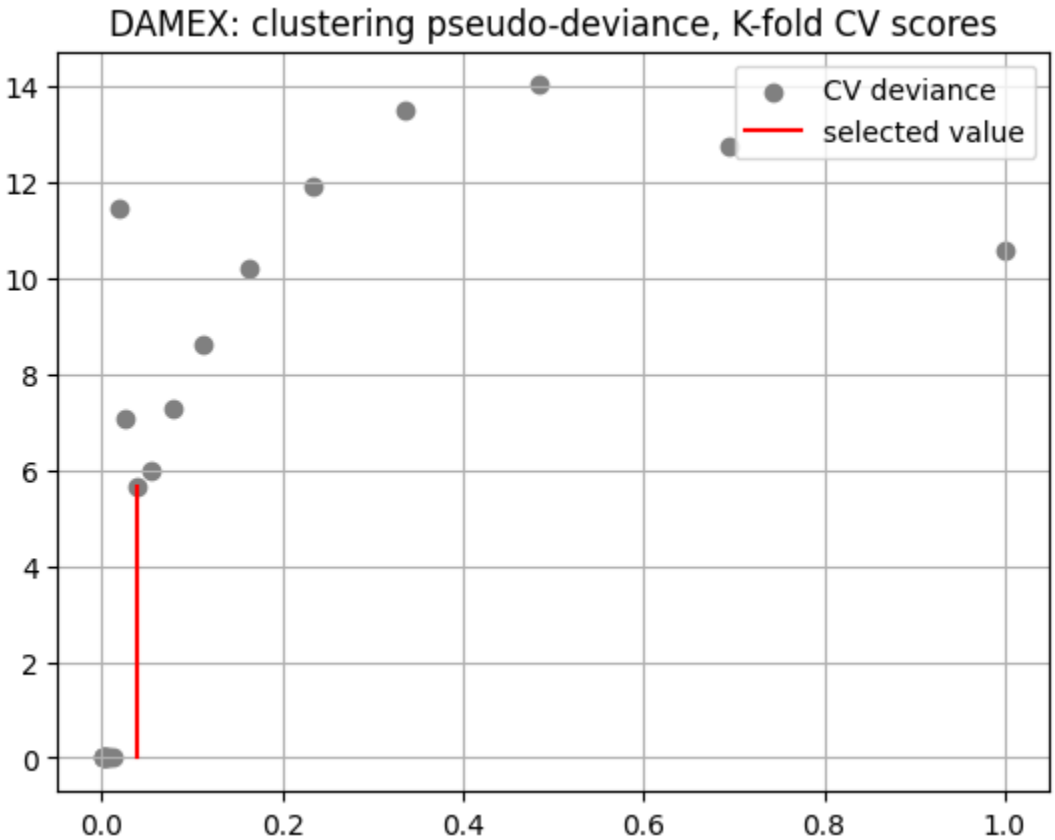
In [ ]:
```
# select epsilon with AIC and update model:
neps = 20
eps_vect = np.geomspace(10**(-3), 1, num=neps)
eps_select_AIC, aic_opt, aic_values = clustering.select_epsilon_AIC(
    eps_vect, X, plot=True, update_epsilon=False)

print('epsilon parameter selection (AIC):')
print(mlx.round_signif(eps_select_AIC, 2))

# select epsilon with CV but don't update model:
eps_select_CV, _, _ = clustering.select_epsilon_CV(eps_vect, X, plot=True,
                                                   update_epsilon=False)
```

```
print('epsilon parameter selection (CV):')
print(mlx.round_signif(eps_select_CV, 2))

# Check that epsilon stored in object is still the default one
clustering.epsilon
```

```
DAMEX: selected epsilon with AIC criterion 0.038
```




```
epsilon parameter selection (AIC):
0.038
DAMEX: selected epsilon with CV criterion 0.038
```

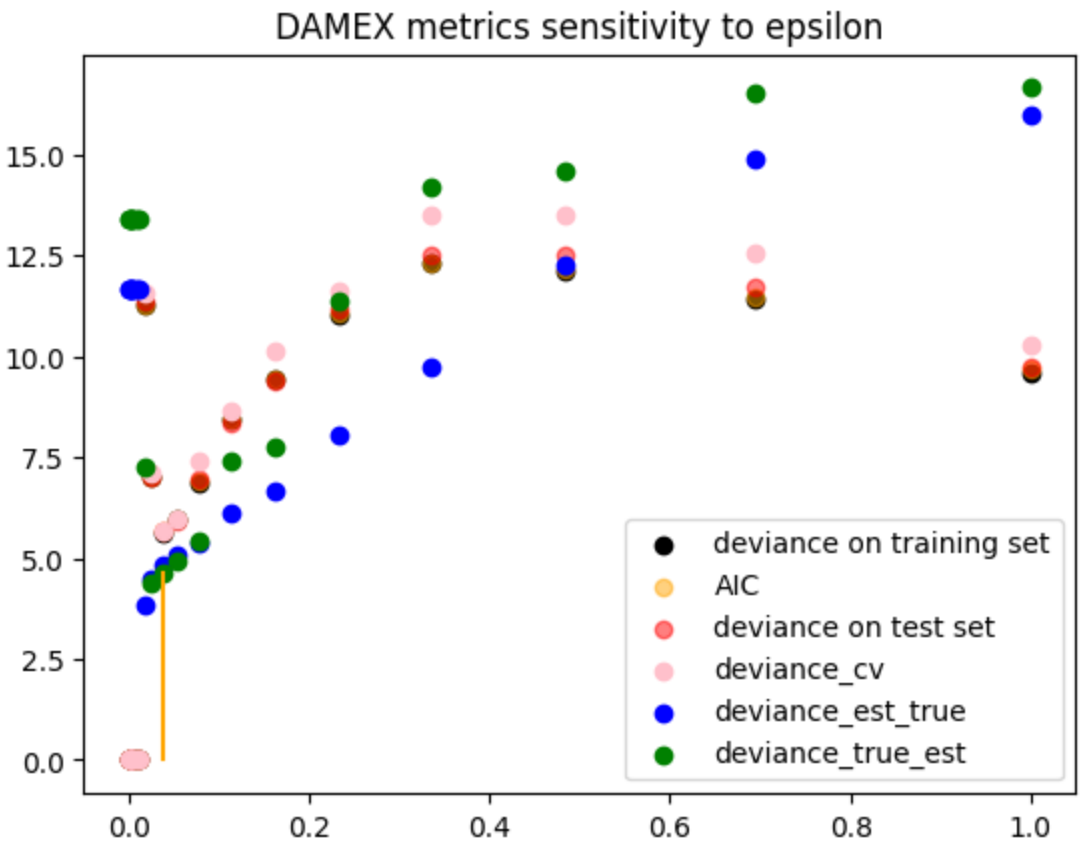


```
epsilon parameter selection (CV):
0.038
```

Out[ ]: False

Wrapping up: Comparison of All Metrics:

In [ ]:
```
# # Agreement with other metrics: recommended choice of epsilon here: by CV
eps_select = eps_select_CV
i_select = np.where(eps_select == eps_vect)[0][0]
aic_vals = np.zeros(neps)
deviance = np.zeros(neps)
deviance_train = np.zeros(neps)
deviance_cv = np.zeros(neps)
deviance_est_true = np.zeros(neps)
deviance_true_est = np.zeros(neps)

for i in range(neps):
    clust = mlx.damex(epsilon=eps_vect[i], min_counts=min_counts,
                      thresh_train=threshold,
                      thresh_test=threshold,
                      include_singletons=include_singletons)
    faces, masses = clust.fit(Xt, standardize=False)
    aic_vals[i] = clust.get_AIC(Xt, standardize=False)
    deviance[i] = clust.deviance(std_Xtest, standardize=False)
    deviance_train[i] = clust.deviance(Xt, standardize=False)
    deviance_cv_scores = clust.deviance_CV(Xt, standardize=False,
                                           random_state=13+137*i, cv=5)
    deviance_cv[i] = np.mean(deviance_cv_scores)
    deviance_est_true[i], deviance_true_est[i] = clust.deviance_to_true(
        faces_true, wei_true)

if True:
    plt.scatter(eps_vect, deviance_train, c='black',
                label='deviance on training set')
    plt.scatter(eps_vect, aic_vals, c='orange', label='AIC', alpha=0.5)
    plt.scatter(eps_vect, deviance, c='red',
                label='deviance on test set', alpha=0.5)
    plt.scatter(eps_vect, deviance_cv, c='pink',
                label='deviance_cv')
    plt.scatter(eps_vect, deviance_est_true, c='blue',
                label='deviance_est_true')
    plt.scatter(eps_vect, deviance_true_est, c='green',
                label='deviance_true_est')
    plt.plot([eps_select, eps_select], [0, deviance_true_est[i_select]],
             c='orange')
    plt.legend()
    plt.title("DAMEX metrics sensitivity to epsilon")
    plt.show()
```



Conclusion: All goes as planned:

- Deviance on the testing set and AIC are very close or even indistinguishable as soon as epsilon is not too small, reflecting the fact that AIC is meant to estimate the deviance on a test set.
- Deviance on the training set and AIC are close because the number of 'parameters' of the model (i.e., number of faces) is small compared with the sample size (here, the number of extremes).
- Deviance on the training set is close to, but less than, the deviance on the test set, reflecting slight overfitting.
- Deviance on the test set is a reasonable approximation of the deviance of the estimated parameter from the true model (not the other way around).
- CV deviance is different from the test deviance but follows a similar pattern.

Take-home message: To choose epsilon, use `select_epsilon_AIC` and `select_epsilon_CV`. The results should be similar. If they are not, you may be in trouble because the (extreme) sample size may be too small.

In [ ]:
```
# subfaces and masses
subfaces_select, masses_select = clustering.fit(Xt, eps_select, False)
weights_select = masses_select / np.sum(masses_select)
```

```
dict_faces_select, dict_weights_select = mlx.list_to_dict(subfaces_select,
                                                          weights_select)

print("Damex with selected epsilon: subfaces")
pp.pprint(dict_faces_select)
print("True subfaces")
pp.pprint(faces_dict_true)

print("final output: weights = normalized masses")
pp.pprint(dict_weights_select)
print("True weights")
pp.pprint(mass_dict_true)
```

```
Damex with selected epsilon: subfaces
{1: [],
 2: [[10, 13], [0, 5], [4, 8]],
 3: [[0, 3, 8]],
 4: [],
 5: [[7, 8, 13, 14, 17], [1, 3, 8, 14, 19]],
 6: [[2, 9, 11, 12, 16, 17], [0, 6, 7, 11, 16, 19]],
 7: [],
 8: [],
 9: [[1, 2, 4, 6, 9, 13, 15, 18, 19], [1, 2, 4, 6, 7, 9, 13, 15, 18]],
 10: [[0, 1, 2, 3, 4, 6, 7, 9, 12, 19], [1, 2, 4, 6, 7, 9, 13, 15, 18, 19]]}
True subfaces
{1: [],
 2: [[0, 5], [10, 13], [4, 8]],
 3: [[0, 3, 8]],
 4: [],
 5: [[7, 8, 13, 14, 17], [1, 3, 8, 14, 19]],
 6: [[0, 6, 7, 11, 16, 19], [2, 9, 11, 12, 16, 17]],
 7: [],
 8: [],
 9: [],
 10: [[0, 1, 2, 3, 4, 6, 7, 9, 12, 19], [1, 2, 4, 6, 7, 9, 13, 15, 18, 19]]}
final output: weights = normalized masses
{1: [],
 2: [0.11, 0.11, 0.065],
 3: [0.07],
 4: [],
 5: [0.12, 0.096],
 6: [0.14, 0.082],
 7: [],
 8: [],
 9: [0.023, 0.019],
 10: [0.088, 0.075]}
True weights
{1: [],
 2: [0.062, 0.067, 0.029],
 3: [0.042],
 4: [],
 5: [0.092, 0.083],
 6: [0.1, 0.13],
 7: [],
 8: [],
 9: [],
 10: [0.16, 0.22]}
```